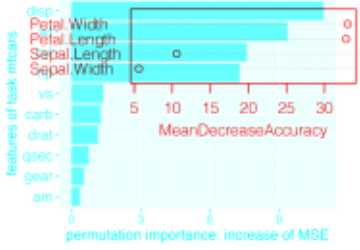
Random Forests: Feature Importance

Feature Importance



Learning goals

- Understand that the goal of defining variable importance is to enhance interpretability of the random forest
- Know definition of variable importance based on improvement in split criterion
- Know definition of variable importance based on permutations of OOB observations

Learning goals
Understand that the goal of feature importance is to enhance interpretability of RF
Know definition of feature permutation
Understand based on improvement in splits

VARIABLE IMPORTANCE

RFs improve accuracy by aggregating multiple decision trees but **lose interpretability** compared to a single tree. **Feature importance** mitigates this problem.

- How much does performance decrease, if feature is removed / rendered useless?
- We perturb values of considered feature
- Removes association between feature and target, keeps marginal distribution
- Compute VI as OOB (without and with permuted features) by predicting OOB data to **efficiently compute FI during training**
- Avoid: not only overfitting (effect would be screened), but one should use "OOB test data" during training

- Single trees are highly interpretable
- Random forests as ensembles of trees lose this feature
- Contributions of the different features to the model are difficult to evaluate
- Way out: variable importance measures
- Basic idea: by how much would the performance of the random forest decrease if a specific feature were removed or rendered useless?

ID	Color	Form	Origin	Length	Banana
1	yellow	round	domestic	10	no
2	brown	oblong	imported	11	yes
3	green	oblong	imported	12	yes
4	yellow	oblong	domestic	14	yes

VARIABLE IMPORTANCE 2

Measure based on improvement in split criterion

for features x_j, $j = 1$ to p **do**

  for tree base learners $\hat{b}^{[m]}$, $m = 1$ to M **do**

    Find all nodes $\mathcal{N}$ in $\hat{b}^{[m]}$ that use x_j.

    Compute improvement in splitting criterion achieved by them.

    Add up these improvements.

  end for

  Add up improvements over all trees to get feature importance of x_j.

end for

VARIABLE IMPORTANCE / 3

Alternative: Add up all *improvements* in splits where feature x_j is used.

Measure based on permutations of OOB obs.

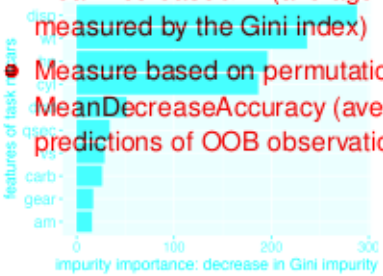
Estimate OOB error $\widehat{\text{err}}_{\text{OOB}}$.
for features $x_j, j = 1$ to p **do**
  Perform permutation ψ_j on x_j to distort feature-target relation for x_j.
  for distorted observations $(\mathbf{x}^{(i)}_{\psi_j}, y^{(i)}), i = 1$ to n **do**
    Compute OOB prediction $\hat{y}^{(i)}_{\text{OOB},\psi_j}$.
    Compute corresponding loss $L(y^{(i)}, \hat{y}^{(i)}_{\text{OOB},\psi_j})$.
  end for
  Estimate importance of j-th variable
  $\widehat{\text{VI}}_j = \widehat{\text{err}}_{\text{OOB},\psi_j} - \widehat{\text{err}}_{\text{OOB}}$
  $= \frac{1}{n}\sum^{n} L(y^{(i)}, \hat{y}^{(i)}_{\text{OOB},\psi_j}) - \widehat{\text{err}}_{\text{OOB}}$.
end for

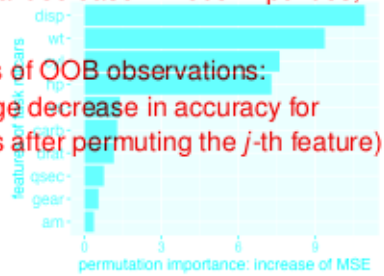
1: **for** features $x_j, j = 1 \to p$ **do**
2: **for** all models $\hat{b}^{[m]}, m = 1$ to M **do**
3: Find all splits in $\hat{b}^{[m]}$ on x_j
4: Extract improvement / risk reduction for these splits
5: Sum them up
6: **end for**
7: Add up improvements over all trees for FI of x_j
8: **end for**

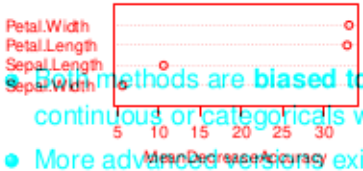
VARIABLE IMPORTANCE / 4

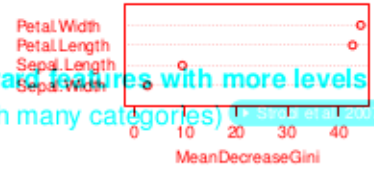
Let's [illegible]

- Measure based on improvement in split criterion: MeanDecreaseGini (average total decrease in node impurities, measured by the Gini index)
- Measure based on permutations of OOB observations: MeanDecreaseAccuracy (average decrease in accuracy for predictions of OOB observations after permuting the j-th feature)









- Both methods are **biased toward features with more levels** (i.e., continuous or categoricals with many categories)
- More advanced versions exist
- PFI and FI have been generalized, see our lecture on IML!